ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT I

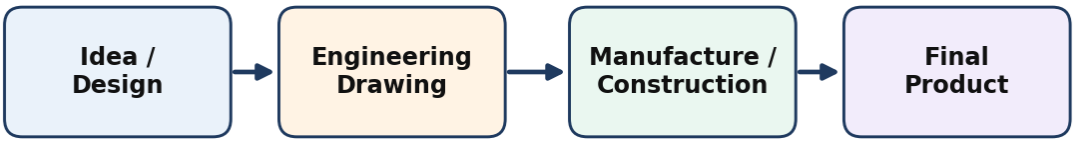
Drawing Basics, Curves & Scales

Learn the alphabet of the language of engineers.

SYLLABUS COVERED

- ▸ Lines, Lettering & Dimensioning
- ▸ Geometrical constructions
- ▸ Regular polygons – general methods
- ▸ Ellipse, Parabola, Hyperbola (general method)
- ▸ Cycloids & Involutes
- ▸ Normal and tangent to curves
- ▸ Plain, Diagonal & Vernier scales

Engineering drawing is the universal language of engineers



Lines • Lettering • Dimensions • Views • Sections

Fig. 0.1 — Why we study Engineering Graphics

1. Introduction & Drawing Essentials

DEFINITION

Engineering Drawing is a graphical language that communicates the **shape, size, material and manufacturing details** of an object accurately and unambiguously, following standard rules (**BIS – SP 46:2003**).

Instrument	Use	Tip
Drawing board + Mini-drafter / T-square	Fix sheet; draw horizontal & inclined lines	Keep head firmly on board edge
Set squares (45° & 30°–60°)	Vertical lines; angles 15°, 30°, 45°, 60°, 75°, 90°	Combine both to get 15° steps
Compass & divider	Draw circles / arcs; transfer & divide lengths	Needle slightly longer than lead
Pencils H, 2H, HB	2H – construction; H – dimension/centre; HB – outlines & lettering	Sharp conical point
French curves, protractor, eraser	Draw smooth curves; measure angles; clean up	Join plotted points freely

Sheet	A0	A1	A2	A3	A4
Size (mm)	841 × 1189	594 × 841	420 × 594	297 × 420	210 × 297

Sheet rule: each next size is obtained by folding the previous size in half (A_{n+1} = ½ A_n); aspect ratio 1 : √2. Every sheet has a **border**, a **title block** (bottom-right) and a **first/third-angle symbol**.

2. Types of Lines (BIS)

Type	Sample line	Name	Application
A	———	Continuous thick	Visible outlines, visible edges
B	———	Continuous thin	Dimension, extension, leader, projection lines; hatching
C	~~~	Continuous thin, freehand	Short break lines
D	—/\/—	Continuous thin with zig-zag	Long break lines
E	- - - - - -	Dashed (thin / medium)	Hidden outlines & hidden edges
F	—·—·—·—	Chain thin (long-dash dot)	Centre lines, lines of symmetry, locus
G	▬·—·—·▬	Chain thin, thick at ends	Cutting-plane lines
H	—··—··—	Chain thin double-dash	Adjacent parts, alternate positions

Fig. 2.1 — Types of lines and where they are used

KEY POINTS

- **Priority** when lines coincide: visible outline → hidden line → cutting plane → centre line → dimension line.
- Thick : thin line ratio ≈ **2 : 1** (e.g. 0.7 mm : 0.35 mm).
- Centre lines extend **~3 mm** beyond the outline and end with a long dash; hidden lines meet at corners with dashes.

3. Lettering

Lettering gives **names, notes and figures** on a drawing. It must be **uniform, legible, single-stroke** and fast to write. Two styles exist: **vertical** and **inclined (75°)**; all capitals are recommended.

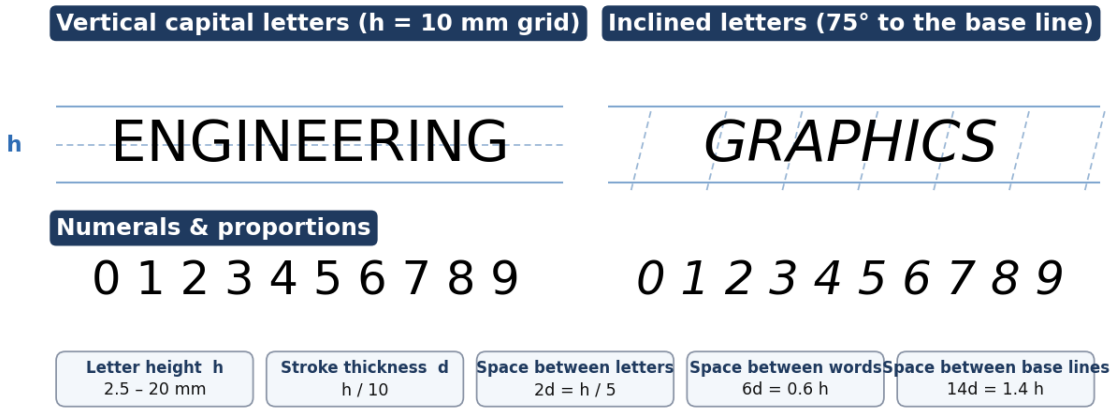


Fig. 3.1 — Single-stroke lettering and standard proportions (Type B: h = 10 d)

Use on drawing	Typical letter height (h)
Main titles / drawing number	10 – 14 mm
Sub-titles, headings	5 – 7 mm
Notes, dimension figures, tolerances	2.5 – 3.5 mm

4. Dimensioning

DEFINITION

Dimensioning is the process of giving **size and location information** on a drawing so that the part can be manufactured without any further questions.

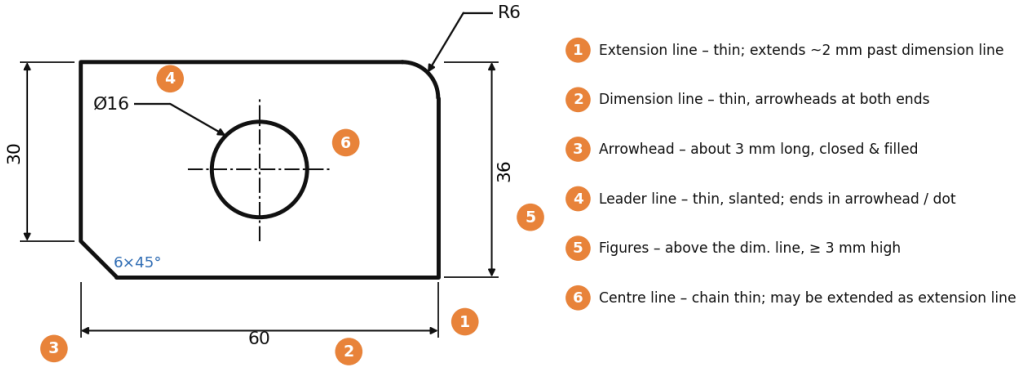


Fig. 4.1 — Elements of dimensioning

Golden rules

- Place dimensions **outside** the view; avoid crossing dimension lines; keep the **smaller dimension nearest** the outline.
- Give each dimension **only once**; never use a centre line, outline or extension line as a dimension line.
- Use symbols: **Ø** diameter, **R** radius, □ square, **SØ / SR** sphere; all in **mm** without writing the unit.
- Extension lines are drawn ⊥ to the feature, ~**2 mm beyond** the dimension line, with a ~1 mm gap from the outline.

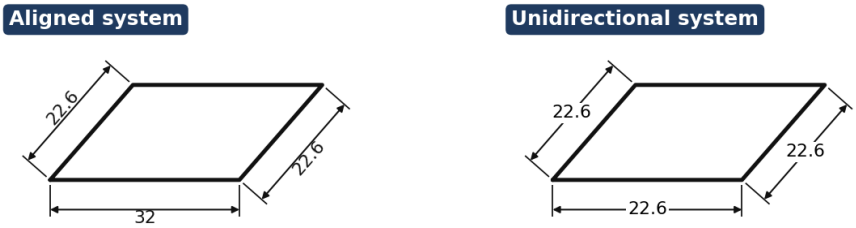
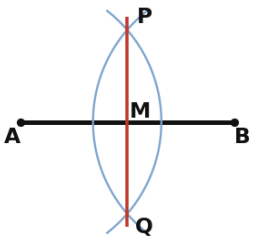


Fig. 4.2 — Aligned and unidirectional systems of placing figures

System	Rule for figures	Where used
Aligned	Parallel to dim. line, readable from bottom or right	Most general drawings
Unidirectional	All figures horizontal; dim. line broken at middle	Large drawings, aircraft/automobile

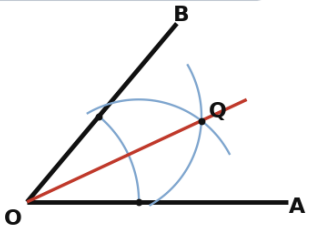
5. Geometrical Constructions

(a) Bisect a line AB



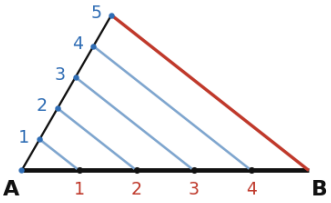
AM = MB, PQ ⊥ AB

(b) Bisect angle AOB



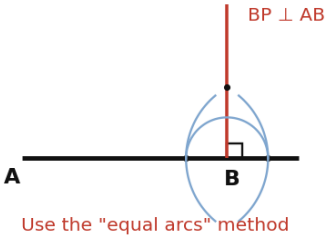
∠AOQ = ∠QOB

(c) Divide AB into 5 equal parts



Join 5–B, then draw ∥ lines

(d) Perpendicular at end B of AB



Use the "equal arcs" method

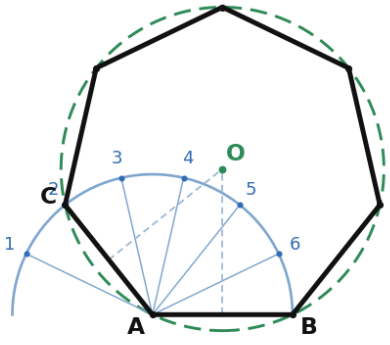
Fig. 5.1 — Basic geometrical constructions

Angle by set squares	Combination
30°, 60°, 90°	30°–60° set square
45°	45° set square
15° / 75°	45° and 30°–60° together (45° − 30° / 45° + 30°)
105° / 120°	45° + 60° / 60° + 60°

6. Regular Polygons – General Methods

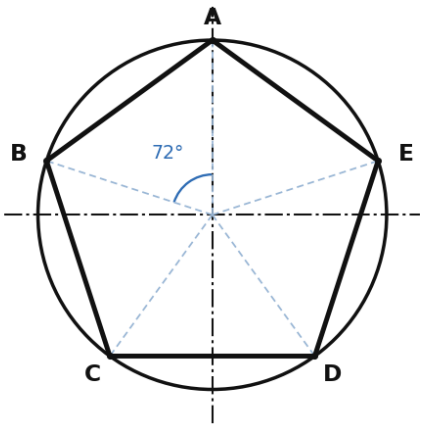
A **regular polygon** has all sides and all interior angles equal; all its vertices lie on a circle. **General method** = one construction that works for *any* number of sides n.

(a) Heptagon on a given side AB



Semicircle on AB ÷ 7 → join A–2 (from far end)

(b) Polygon inside a circle (n = 5)



chord a = 2R·sin(180°/n)

central angle = 360°/n

Fig. 6.1 — General method for a polygon on a given side, and polygon inside a circle

PROCEDURE

1. Draw the given side **AB**. With centre A and radius AB draw a **semicircle** starting from B.
2. Divide the semicircle into **n equal parts** (n = number of sides) and number the points 1, 2, 3 … from the end **far from B**.
3. Join **A–2**; this is the direction of the second side. Mark **AC = AB** on it.
4. Draw perpendicular bisectors of AB and AC – they meet at the **centre O** of the polygon.
5. With O as centre and OA as radius draw the **circumscribing circle** and step off AB around it. Join the points.

FORMULAS

Interior angle = **(n − 2) × 180° / n** | Exterior angle = **360° / n** | Side a = **2R sin(180°/n)** | Inradius r = **a / (2 tan(180°/n))**

Area = **n a² / (4 tan(180°/n))** | Number of diagonals = **n(n − 3)/2**

Sides n	3	4	5	6	7	8
Name	Triangle	Square	Pentagon	Hexagon	Heptagon	Octagon
Interior angle	60°	90°	108°	120°	128.57°	135°
a / R	1.732	1.414	1.176	1.000	0.868	0.765

7. Engineering Curves – Conic Sections

DEFINITION

A **conic** is the locus of a point P that moves so that the ratio of its distance from a fixed point **(focus F)** to its distance from a fixed straight line **(directrix D)** is constant. This ratio is the **eccentricity** e = PF / PN.

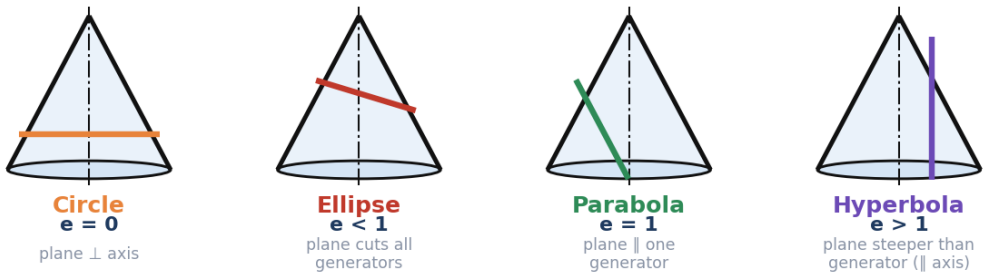


Fig. 7.1 — Conics are plane sections of a cone

Curve	e	Standard equation	Real-life use
Ellipse	< 1	$x^2/a^2 + y^2/b^2 = 1$	Planet orbits, elliptical arches, boiler manholes
Parabola	= 1	$y^2 = 4ax$	Headlight / antenna reflectors, suspension-bridge cables
Hyperbola	> 1	$x^2/a^2 - y^2/b^2 = 1$	Cooling towers, sonic-boom curves, LORAN navigation
Rect. hyperbola	= √2	$x \cdot y = c$	Boyle's law PV = constant

6.1 Ellipse

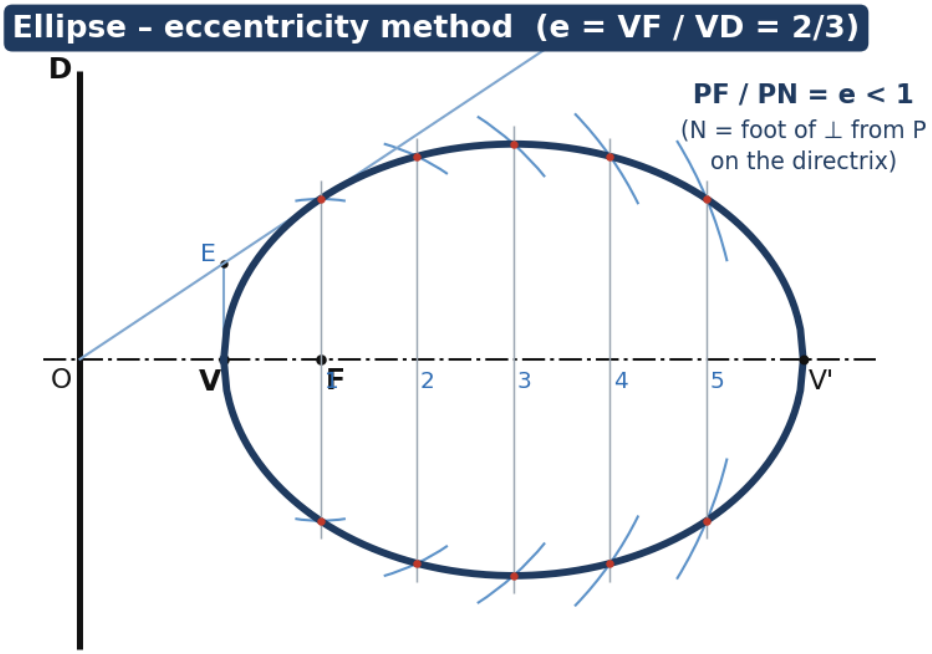


Fig. 7.2 — Ellipse by the general (eccentricity) method

PROCEDURE

1. Draw directrix **DD** and the axis ⊥ to it, meeting at **O**. Mark focus **F** on the axis (e.g. OF = 40 mm).
2. For e = 2/3 divide OF into **2 + 3 = 5** equal parts; vertex **V** is at the 3rd part from O (VF/VO = 2/3).
3. At V erect **VE = VF** and join **O–E** (slope = e). Mark points 1, 2, 3 … on the axis and draw verticals.
4. With **F as centre and radius = (vertical to line OE)** cut the vertical above & below the axis to get points on the curve.
5. Join the points with a **French curve**. The second vertex V′ is found by the same rule on the other side of F.

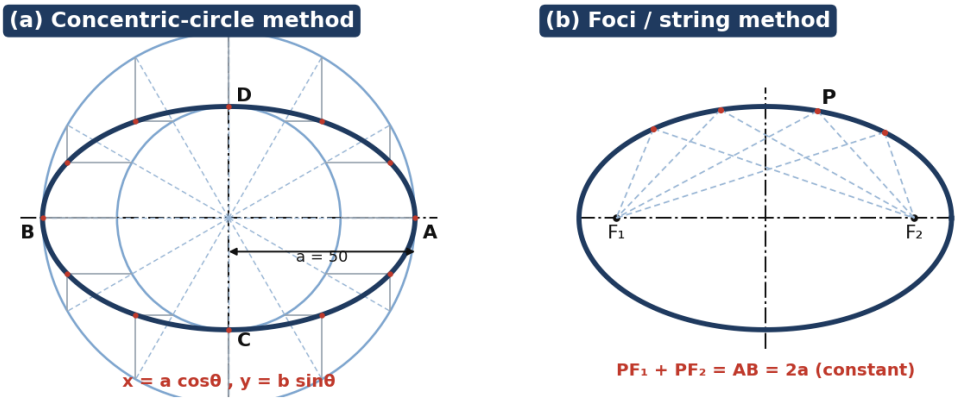


Fig. 7.3 — Other methods: concentric circles & foci (string) method

FORMULAS

Eccentricity **e = √(1 − b²/a²)** | Foci at **±ae** from centre | Sum of focal distances **PF₁ + PF₂ = 2a** | Major axis 2a, minor axis 2b

6.2 Parabola

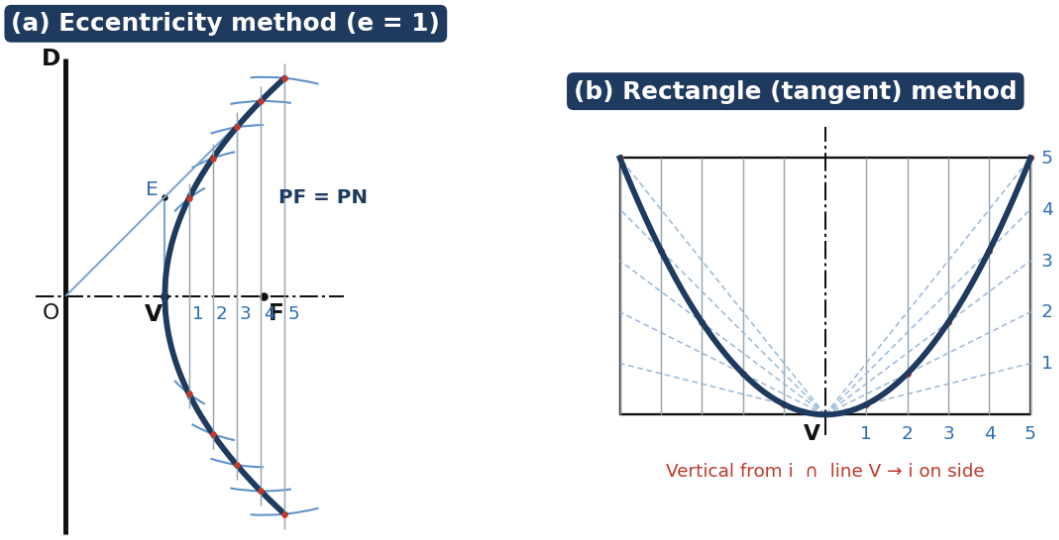


Fig. 7.4 — Parabola – eccentricity method and rectangle method

Eccentricity method: same as the ellipse, but e = 1, so the vertex V is the **mid-point of OF** and VE = VF = VO. Radius for every point = distance of the vertical from the directrix (PF = PN). **Rectangle method:** use when the base and axis lengths are given; divide the half-base and the axis into the same number of equal parts and join the vertex to the division points.

6.3 Hyperbola

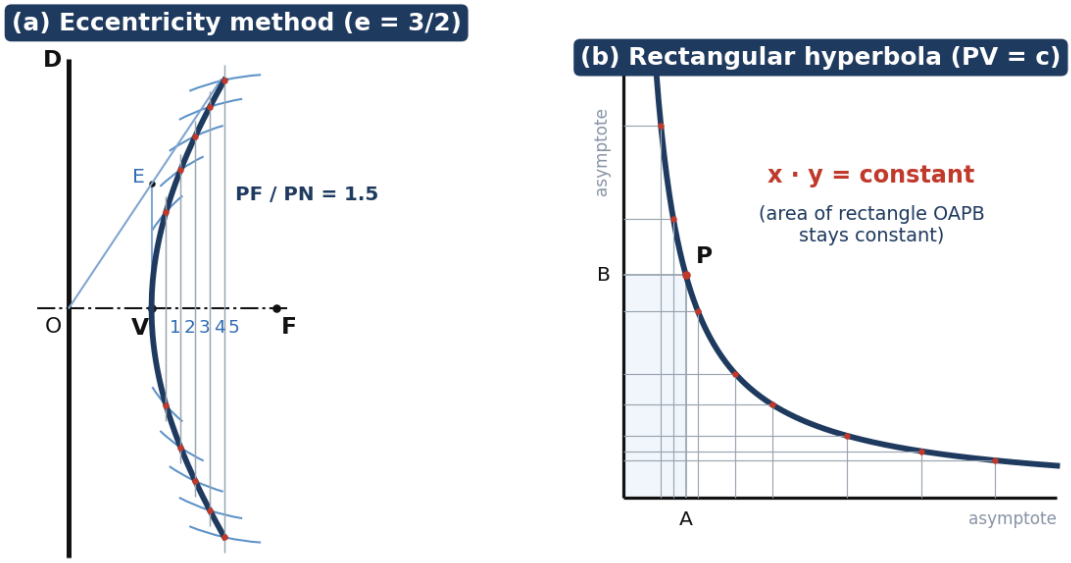


Fig. 7.5 — Hyperbola – eccentricity method and rectangular hyperbola

Eccentricity method: for e = 3/2 divide OF into **2 + 3** parts so that VO : VF = 2 : 3 (VF > VO). Points are found by the same “vertical + arc from F” step and the curve opens away from the directrix. **Rectangular hyperbola:** asymptotes are at 90°; draw rectangle OAPB through P and, through any point on PA/PB lines, the intersection of the vertical and horizontal gives points of the curve (x · y = constant).

8. Cycloidal Curves

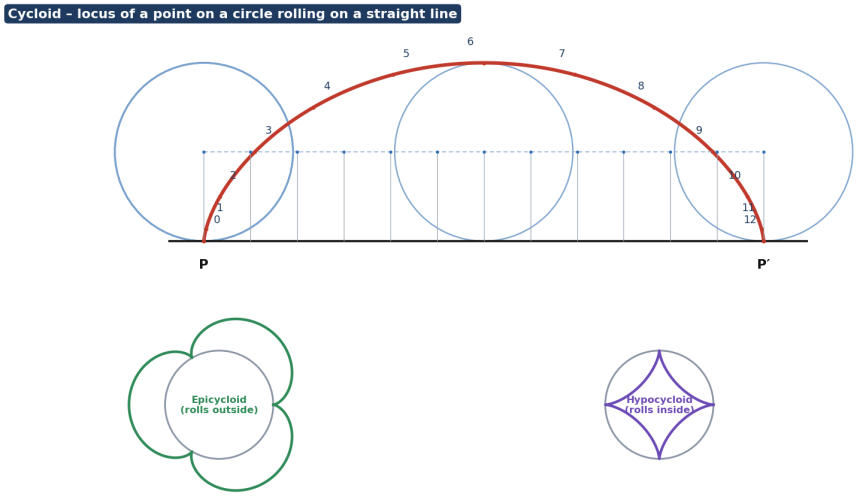


Fig. 8.1 — Cycloid, epicycloid and hypocycloid

Curve	Generating circle rolls …	Directing line / circle
Cycloid	on a straight line (without slipping)	Straight line
Epicycloid	on the **outside** of another circle	Directing circle (radius R)
Hypocycloid	on the **inside** of another circle	Directing circle (radius R)

PROCEDURE

1. Draw the generating circle (radius r) and a base line of length **πD = 2πr**; divide both into **12 equal parts**.
2. Draw the line of centres parallel to the base and locate the successive centres C_1, C_2 … C_{12} by verticals.
3. Through each division of the circle draw a **horizontal line**. With centre C_k and radius r cut horizontal k at $\mathbf{P_k}$.
4. Join P_0, P_1 … P_{12} by a smooth curve.

FORMULAS

Cycloid: **x = r(θ − sin θ), y = r(1 − cos θ)** | Base length of one arch = **2πr** | Height = **2r** | Arc length = **8r**

Epi/Hypocycloid: angle subtended at centre of directing circle **θ = (r / R) × 360°**

9. Involute

DEFINITION

An **involute** is the locus of the free end of a **taut string** unwound from a circle (or polygon). At any point, the **normal to the involute is tangent to the base circle**.

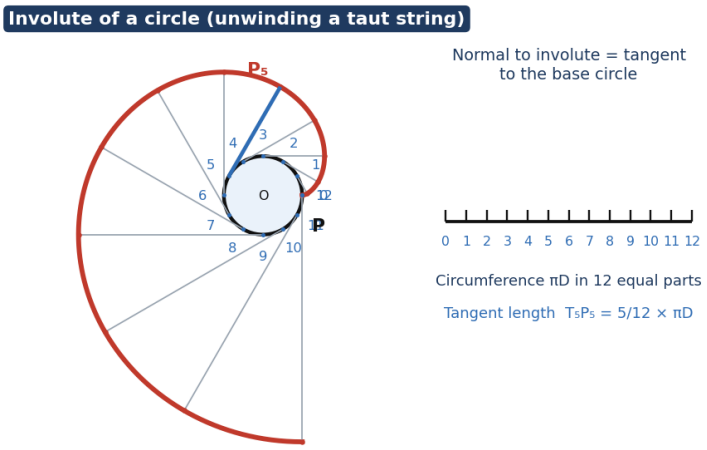


Fig. 9.1 — Involute of a circle

PROCEDURE

1. Draw the base circle (radius r) and divide its circumference into **12 equal parts**; number them 1 … 12.
2. At each point draw the **tangent** (⊥ to the radius) in the same sense.
3. Make tangent at k equal to **k/12 of the circumference** (k × πD/12) – i.e. the length of the arc P–k.
4. Join the end-points by a smooth curve.

APPLICATIONS

Involute profiles are used for **gear teeth** – they give constant velocity ratio and smooth meshing. Cycloids are used in clock/watch gears and pump rotors.

10. Normal & Tangent to Curves

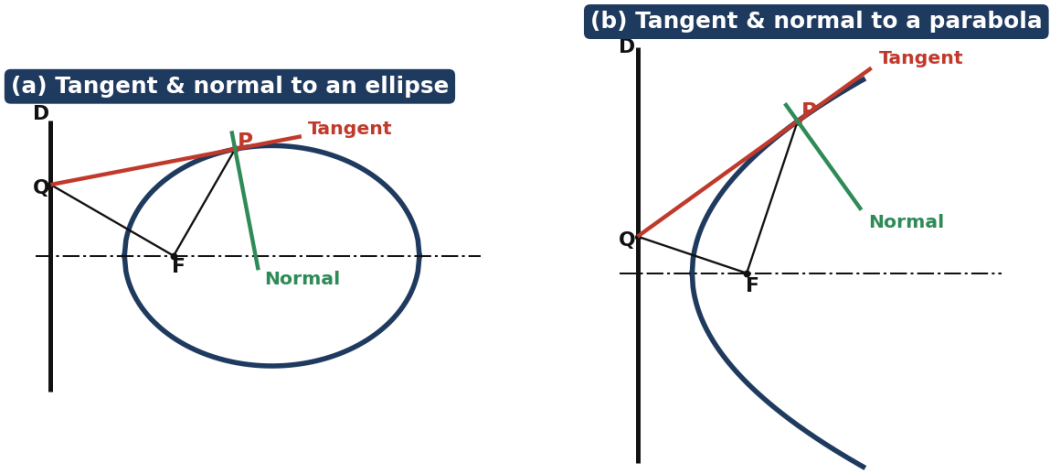


Fig. 10.1 — Tangent and normal using the focus–directrix property

PROCEDURE

Conics (ellipse, parabola, hyperbola) at point P:

1. Join **P to the focus F**.
2. At F draw a line **perpendicular to PF** to meet the directrix at **Q**.
3. Join **QP** and extend – this is the **tangent**.
4. Draw a line **⊥ to the tangent at P** – this is the **normal**.

Curve	Normal at a point P
Circle	Passes through the centre
Involute	Tangent to the base circle
Cycloid	Joins P to the point where the generating circle (through P) touches the base line
Any curve	Draw a circle at P, mark where it cuts the curve on both sides; their chord's ⊥ bisector gives the normal

11. Scales

DEFINITION

A **scale** is a ruler on which actual lengths are shown reduced or enlarged. **Representative Fraction (RF)** = length of drawing / actual length (same units).

FORMULAS

RF = **1 / (ground length per 1 cm of drawing in cm)** | Length of scale = **RF × maximum length to be measured**

Reduced scale RF < 1 (1:50) • **Full** size RF = 1 • **Enlarged** RF > 1 (5:1)

Scale	Reads	Principle
Plain	2 consecutive units (m & dm)	Extra main division at left of 0 divided into sub-divisions
Diagonal	3 units (m, dm, cm)	Similar triangles: diagonals × 10 horizontal lines
Vernier	2–3 units, very fine (least count)	n vernier divisions = (n − 1) main divisions

8.1 Plain scale

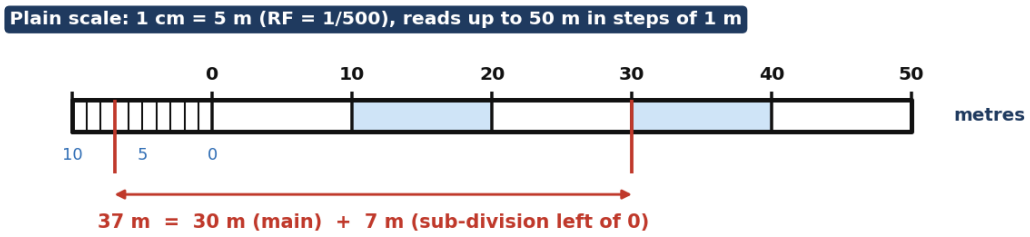


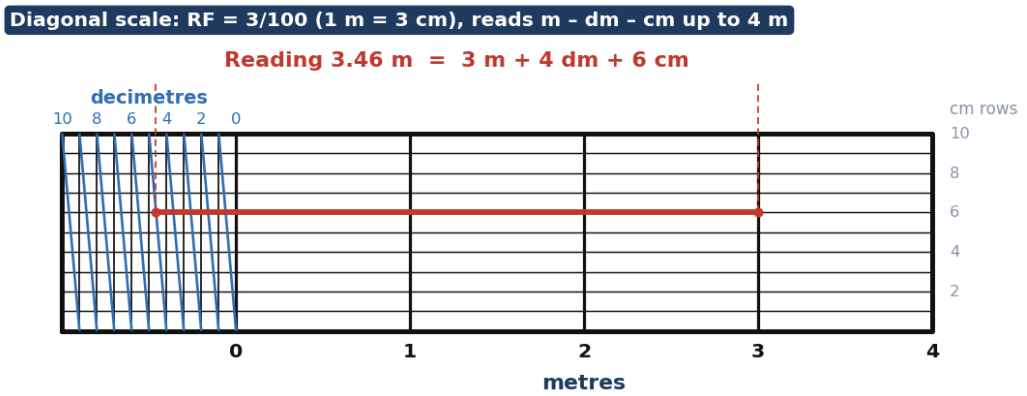
Fig. 11.1 — Plain scale

SOLVED EXAMPLE

Draw a plain scale of 1 cm = 5 m to read up to 50 m; mark 37 m.

RF = 1 cm / 500 cm = **1/500**. Length = 50 m ÷ 5 = **10 cm**. Draw 10 cm, divide into 5 main parts (10 m each = 2 cm). Divide the extra left division into 10 parts (1 m each). Mark 7 m on the left and 30 m on the right: distance = **37 m**.

8.2 Diagonal scale



Each diagonal moves 1 dm-division (3 mm) over 10 rows, so every row adds 1/10 dm = 1 cm.

Fig. 11.2 — Diagonal scale

SOLVED EXAMPLE

Diagonal scale RF = 3/100 to read m, dm, cm up to 4 m; show 3.46 m.

Length = 3/100 × 4 m = **12 cm**. Draw 12 cm, divide into 4 main divisions (3 cm = 1 m). Draw a rectangle of any height, divided into **10** equal rows; subdivide the left division into 10 parts (dm) and draw diagonals. 3.46 m = 3 m (right vertical) + 4 dm (diagonal no. 4) + 6 cm (row no. 6).

8.3 Vernier scale

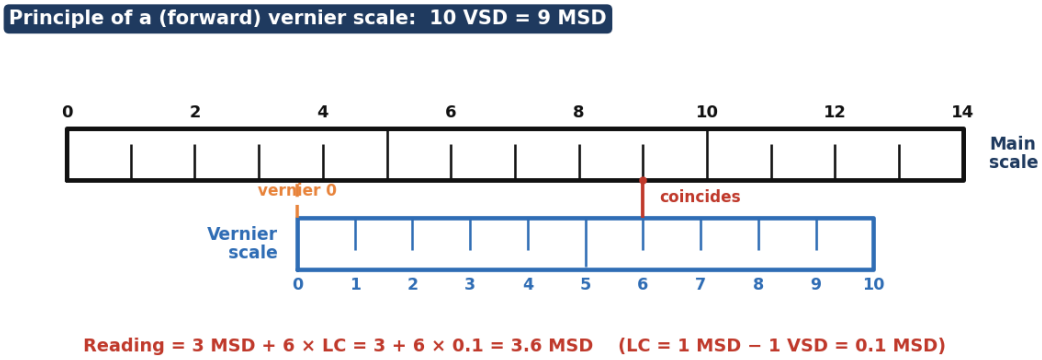


Fig. 11.3 — Vernier scale principle

FORMULAS

Forward vernier: **n VSD = (n − 1) MSD** | Retrograde: **n VSD = (n + 1) MSD** | **Least count LC = 1 MSD − 1 VSD = MSD / n**

Reading = **(main-scale reading) + (coinciding vernier division) × LC**

SOLVED EXAMPLE

Vernier scale RF 1/25 to read 0.01 m on 4 m. 1 m = 4 cm → length = **16 cm**; 1 MSD (0.1 m) = 4 mm; LC = 0.01 m → n = 10; vernier length = 9 MSD = **36 mm** divided into 10 equal parts (3.6 mm each).

12. Quick Revision & Exam Questions

QUICK SUMMARY

- **Lines:** thick – outline; thin – dimension/extension; dashed – hidden; chain – centre/cutting plane.
- **Lettering:** single stroke, h = 10d, inclined at 75°, spacing 2d.
- **Polygons:** semicircle ÷ n → join A–2; interior angle = (n−2)·180/n.
- **Conics:** e<1 ellipse, e=1 parabola, e>1 hyperbola; PF = e·PN; tangent via perpendicular at F.
- **Cycloid** – circle rolling on a line (base = πD); **involute** – string unwound (tangent = arc length).
- **Scales:** RF = drawing/actual; length = RF × max length; LC = MSD/n.

EXAM-ORIENTED QUESTIONS

Q1. List the types of lines used in engineering drawing with one application each. **[2]**

Q2. State the principles of dimensioning and explain aligned and unidirectional systems. **[4]**

Q3. Construct a regular heptagon of side 30 mm by the general method. **[6]**

Q4. Draw an ellipse by the eccentricity method when the distance of focus from directrix is 50 mm and e = 3/4. **[8]**

Q5. Construct a parabola with focus 40 mm from the directrix. Draw tangent and normal at a point 50 mm from the focus. **[8]**

Q6. Draw a rectangular hyperbola when a point P is 20 mm and 40 mm from the two asymptotes. **[8]**

Q7. Draw a cycloid for a circle of 50 mm diameter rolling on a straight line. Draw the normal at a point 40 mm above the base. **[8]**

Q8. Draw an involute of a circle of 40 mm diameter; draw a tangent and normal at a point 100 mm from the centre. **[8]**

Q9. Construct a diagonal scale of RF = 1/40 to read metres, decimetres and centimetres up to 5 m. Show 3.57 m. **[6]**

Q10. Draw a vernier scale of RF 1/20 to read up to 3 m with a least count of 0.01 m. **[6]**